

Session 2: HTML and CSS

Web Development Fundamentals

What are HTML & CSS?

HTML

Structure of the website

CSS

Design and styling of the website

Setup

1 Step 1 — No Installation Needed

HTML and CSS come built into every browser — Chrome, Brave, Firefox, etc. No downloads or installs required to get started.

2 Step 2 — Install Live Server (VS Code Extension)

Install the **Live Server** extension in VS Code. It saves time by automatically reloading your website in the browser every time you save a file.

3 Step 3 — Create Your Project Files

Create two files in your project folder:

- `index.html` — your webpage structure
- `style.css` — your design and styles

4 Step 4 — Boilerplate & Link CSS

In `index.html`, press the shortcut `! then Tab` to auto-generate the HTML boilerplate. Then link your CSS file inside the `<head>` and start building!

Key CSS Properties

Property	Description	Example
color	Sets text colour	<code>color: red;</code>
background-color	Sets background colour	<code>background-color: #fff;</code>
font-size	Sets the text size	<code>font-size: 16px;</code>

font-family	Sets the typeface	font-family: Arial, sans-serif;
font-weight	Bold / normal weight	font-weight: bold;
margin	Space outside the element	margin: 10px;
padding	Space inside the element	padding: 10px;
border	Adds a border	border: 1px solid #ccc;
width / height	Sets element dimensions	width: 100%; height: 50px;
display	Block, inline, flex, grid...	display: flex;
flexbox	1D layout (row or column)	display: flex;
grid	2D layout system	display: grid;
position	static / relative / absolute	position: relative;
text-align	Aligns text	text-align: center;
border-radius	Rounds corners	border-radius: 8px;
box-shadow	Adds shadow to element	box-shadow: 0 2px 8px rgba(0,0,0,.1);
opacity	Sets transparency (0–1)	opacity: 0.8;
transition	Smooth property change	transition: all 0.3s ease;
overflow	Controls overflowing content	overflow: hidden;

Essential HTML Tags

Tag	Purpose
<code><!DOCTYPE html></code>	Declares document as HTML5
<code><html></code>	Root element of the page
<code><head></code>	Metadata (title, links, scripts)
<code><title></code>	Page title shown in browser tab
<code><body></code>	All visible page content
<code><h1> - <h6></code>	Headings (h1 = largest)
<code><p></code>	Paragraph of text
<code></code>	Hyperlink / anchor

<code></code>	Embeds an image
<code><div></code>	Block-level container
<code></code>	Inline container
<code> / / </code>	Unordered / ordered list and items
<code><input></code>	Form input field
<code><button></code>	Clickable button
<code><form></code>	Groups form elements
<code><nav></code>	Navigation section
<code><header> / <footer></code>	Page header / footer
<code><section> / <article></code>	Semantic content sections
<code><link rel='stylesheet'></code>	Links an external CSS file

Session 2: Resources & Tasks

Tools, Inspiration & Practice Assignments

Fonts & Icons

Google Fonts

<https://fonts.google.com/>

Font Awesome (Icons)

<https://fontawesome.com/>

Images

Unsplash

<https://unsplash.com/>

Pexels

<https://pexels.com/>

Animated Icons

Lordicon

<https://lordicon.com/>

UI Design

Figma

<https://figma.com/>

- Create UI mockups
- Use Figma AI for design ideas
- Plan website layouts before coding

CSS Inspiration

CodePen — <https://codepen.io/>

Search for:

#Buttons #Cards #LoginForms #NavigationBars #HoverEffects #Loaders #LandingPages

Example pen:

<https://codepen.io/nodws/pen/PwbOdME>

Recommended VS Code Extensions

1. **Live Server** — Auto-reloads your browser on every save.
2. **Prettier** — Makes your messy code look beautiful automatically.

Tasks

Task 1 Personal Profile Page

Goal: Learn basic HTML structure + simple CSS styling

Requirements:

- Name heading
- Profile image
- Short bio (3–5 lines)
- List of skills (HTML, CSS, etc.)
- Contact section (email / social links)

Focus:

- Proper use of tags: `h1`, `p`, `img`, `ul`, `a`, `div`
- Basic styling: font, color, spacing

➤ **Finally push it to GitHub**

Task 2 Create Your Own Google Clone

Goal: Learn by your own ideas — create your most beautiful, styled Google Homepage clone

Requirements:

- Google logo or your modified Google (use your own font or an AI-generated modified logo)
- Search input field
- Other features that make it look like Google
- Add background images and anything to match your vibe
- No need to make the search input functional **(We will get it working after learning JS!)**